Perfect Copying of Basis States

1 Exercise

Find a machine C that perfectly copies $|0\rangle$ and $|1\rangle$. That is, such that:

$$C(|0\rangle \otimes |0\rangle) = |0\rangle \otimes |0\rangle, \tag{1}$$
$$C(|1\rangle \otimes |0\rangle) = |1\rangle \otimes |1\rangle. \tag{2}$$

Here, the first qubit is the one to be copied, and the second qubit is a blank ancilla.
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2 Step 1: Identifying the Cloning Machine

We require a quantum operation that satisfies:

$$\begin{cases} |0\rangle |0\rangle \mapsto |0\rangle |0\rangle, \\ |1\rangle |0\rangle \mapsto |1\rangle |1\rangle. \end{cases}$$

A **controlled-NOT (CNOT) gate** achieves exactly this behavior. Recall:

$$\text{CNOT}: |c\rangle |t\rangle \mapsto |c\rangle |t \oplus c\rangle, \tag{3}$$

where c is the control qubit (input to copy), t is the target qubit (blank), and $\oplus$ is addition modulo 2.
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3 Step 2: Verification for Basis States

Case 1: $|0\rangle \otimes |0\rangle$

$$\text{CNOT}(|0\rangle |0\rangle) = |0\rangle |0 \oplus 0\rangle = |0\rangle |0\rangle.$$

Case 2: $|1\rangle \otimes |0\rangle$

$$\text{CNOT}(|1\rangle |0\rangle) = |1\rangle |0 \oplus 1\rangle = |1\rangle |1\rangle.$$

Both cases satisfy the required cloning conditions.
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4 Step 3: Limitation

If we attempt to clone a **superposition state**:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle \,,$$

then the output of CNOT is:

$$\mathrm{CNOT}(|\psi\rangle \otimes |0\rangle) = \alpha |0\rangle |0\rangle + \beta |1\rangle |1\rangle \neq |\psi\rangle \otimes |\psi\rangle \,.$$

This illustrates the **No-Cloning Theorem**: perfect cloning of arbitrary unknown states is impossible.

The CNOT gate only perfectly clones **basis states**.

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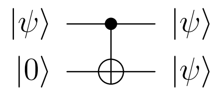
5 Step 4: Quantum Circuit Diagram

The CNOT gate as the cloning machine can be drawn as follows:



- Top wire: original qubit to be cloned ($|0\rangle$ or $|1\rangle$).
- Bottom wire: blank qubit (ancilla).
- $*! < 0em, .025em > - = - < .2em > \bullet -!Ch!Ch[1,0]-!Ch!Ch[0,-1]$: control qubit of CNOT.
- $*+ < .02em, .02em > \bigoplus -!Ch!Ch[0,-1]$: target qubit (receives the copy).

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6 Step 5: Summary

- The machine C that perfectly copies $|0\rangle$ and $|1\rangle$ is the **CNOT gate**.
- Works only for **orthogonal basis states**.
- For arbitrary superpositions, perfect cloning is impossible due to linearity of quantum mechanics.